**Mechatronic Systems Design – Exercise 1**

1. Blink LED connected to pin 13
2. Blink 3 LEDs (11,12,13) alternatively
3. Read the switch connected to pin 7 and use LED in pin 13 to indicate the status of the switch
4. Use 3 switches (7,8,9) to turn on/off the 3 LEDs
5. Vary the intensity of R, G and B individually in RGB LED connected in pins 3, 5 and 6 (PWM pins)
6. Combine various intensities of R, G and B and get different colours
7. Print a text in serial monitor
8. Connect and read a potentiometer in pin. A2 and display the read value in serial monitor (use bread board to connect potentiometer to induino)
9. Vary the intensity of R in RGB LED in pin. 3 using a potentiometer in pin A2
10. Display a text in LCD (use bread board to connect LCD to Induino according to the connection table shown in class). Connect pin 3 of LCD to GND directly (No contrast adjustment)
11. Read the LDR connected to pin A3 and display it in LCD and serial monitor
12. Set a threshold on the value read from LDR and accordingly turn on/off an LED. Also display the value of LDR and status of LED in different rows of LCD and serial monitor.
13. Connect a servo to pin 5 and sweep between its min and max positions
14. Identify the working range of the servo given to you
15. Control the position of servo using a potentiometer
16. Write a code for 3 bit binary counter using 3 LEDs
17. Write a code to count the number of times a switch is pressed and display it in LCD
18. Write a code to count the number of times your finger crosses just above the LDR, within a known time (5s) and display the count in LCD
19. Write a code to continuously receive a string of format “a***b***c” (* represents numbers) through serial port, extract the numbers from the string, save them in separate variables and display the numbers in LCD
20. Communicate between two Induinos in duplex mode (Two teams combined). Connect Rx and Tx of two Induinos using wires, make one the master and another the slave. Both master and slave should have a LDR and a POT connected with suitable circuits.
    - Master should read LDR and POT values and start sending data in the format “aLDRbPOTc” (LDR – value read from LDR, POT – value read from potentiometer). Slave should wait until master sends data.
    - Once slave receives data, extract LDR and POT values that the master has sent, display them in LCD, read LDR and POT values, make it to a string of same format and send to master. Master should wait until slave sends data.
    - This whole process should continue indefinitely

Note:Create a separate folder for this course and save all the programs separately for evaluation.